Multidistribution Time-Series Prototype Learning for Crop Mapping With Sentinel-1 SAR Imagery

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***Abstract*—Time-series synthetic aperture radar (SAR) offers significant potential for crop mapping due to its all-weather, weather-independent imaging capabilities. Existing crop mapping methods with time-series SAR data have achieved good performance. However, these methods often ignore the phenological diversity of the same crop in time-series data, and their limited performance significantly constrains their potential for application in large-scale crop mapping. To address these issues, this article proposes a prototype-based time-series remote sensing crop mapping framework called multi-distribution time-series prototype learning (MDPL). The framework aims to learn multiple time-series prototypes for the same crop for various phenological distributions of crops, effectively capturing the complex and varied phenological characteristics of crops. Second, a phenological-invariant feature learning module is proposed to enhance the model's generalization capability for large-scale crop mapping. Additionally, a new temporal metric is proposed to capture phenological differences. Experimental results on three benchmark datasets have demonstrated the effectiveness and superiority of MDPL compared with state-of-the-art time-series SAR crop mapping methods.**

***Index Terms*—Crop mapping, generalized learning, multidistribution, prototype learning, time-series synthetic aperture radar (SAR) data.**

I. Introduction

WITH the rapid advancement of earth observation technologies, satellite sensors have been widely employed for crop mapping due to their synoptic view, high-frequency revisit cycles, and high spatiotemporal resolution remote sensing imagery [1], [2]. Compared to single-temporal remote sensing images, time-series images have garnered significant attention in the field of agricultural remote sensing in recent years, as they capture the complete growth phenological processes of crops [3], providing critical information for crop identification.

Currently, two primary remote sensing times-series data utilized for crop mapping are optical remote sensing data and synthetic aperture radar (SAR) data. Optical remote sensing effectively captures vegetation photosynthetic parameters, such as the normalized difference vegetation index (NDVI) [4], providing a clear representation of crop growth status. However, optical remote sensing is susceptible to climatic conditions, resulting in limited availability of observational data in persistently cloudy regions. In contrast, SAR data are not susceptible to weather, enabling reliable earth observation at any time [5]. Moreover, the unique scattering mechanisms of SAR and its dense temporal sampling enable accurate characterization of crop morphologies across various phenological stages. Consequently, time-series SAR data are of significant importance in the field of large-scale crop mapping research.

With the rapid development of deep learning (DL), time-series SAR image-based deep models have achieved certain performance in crop mapping [6]. However, most of the studies in crop mapping tasks treat each crop as an independent distribution and assume that crop phenology is consistent within the same region [7]. The assumption limits its applicability in large-scale crop mapping tasks, as crops often exhibit significant phenological variability [8] due to factors such as meteorological conditions, soil properties, and other environmental variables. The crop "Wheat", "Corn" and "Sorghum" is analyzed as a case study to quantify phenological diversity. As shown in Fig. 1(a), even in the same mapping region, the data distribution of the same crop can vary across different plots due to the extensive coverage. Meanwhile, as shown in Fig. 1(b), the figure visualizes its feature variance across different plots, and the curve trajectories are similar between 0 and 180 days, however, the change trajectories of the curves between 180 and 240 reflect the phenological differences of the same crop. There is also a relationship with the phenology of different crops, as shown in Fig. 1(c), the growth stages of corn and sorghum are similar, making it challenging to differentiate between them. The similarity between different crops and the heterogeneity within the same crop reflect the phenological diversity of crops across large-scale regions. Therefore, considering the phenological diversity within crops is crucial for large-scale crop mapping.

Recently, prototype learning is widely studied in computer vision, with prominent applications in semantic segmentation and image classification [9], [10], [11]. Prototype-guided [12] class-level feature learning has gained attention in remote sensing scene classification to solve the problem of sample diversity. Class prototypes, as abstract embeddings of basic semantic representations, play an active role in enhancing

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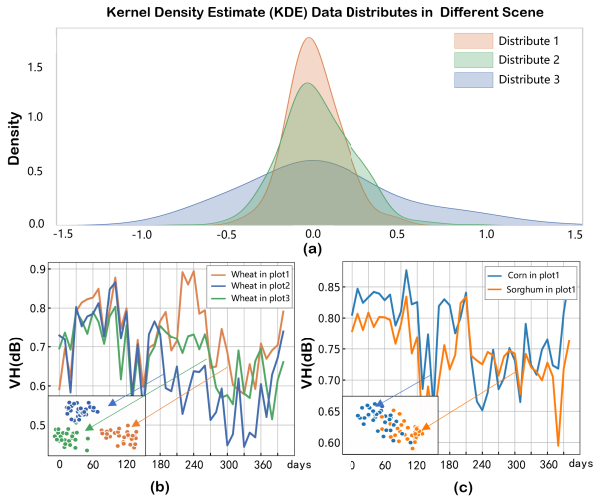

Fig. 1. Phenological diversity illustration. (a) Kernel density estimation of wheat data from different plots in the same region, and different color covers show different distribution states of crops. (b) VH backscatter curves and feature map of wheat in different plots. (c) VH backscatter curves and feature map of corn and sorghum.

intraclass compactness and interclass separability [13]. However, these prototype-based methods exhibit limitations when applied to large-scale crop mapping tasks. First, these methods learn only a single prototype for each category, which is insufficient to capture the phenological diversity of crops in large-scale remote sensing mapping. Second, the lack of metrics to measure time-series phenological features. Third, there is a lack of research on generalization ability, which is essential for its effective application to large-scale land mapping tasks.

To address these issues, this article proposes a novel time-series remote sensing crop mapping framework based on multi-distribution time-series prototype learning (MDPL), aimed at facilitating large-scale crop mapping tasks. In the MDPL framework, we first introduce the fine-grained prototypes generation (FPG) and prototype optimization (POP) modules to model the feature space, aiming to accurately characterize the phenological diversity relationships of crops. The FPG module aims to capture the potential phenological characteristics of crops by generating multiple fine-grained prototypes for each crop. The POP module aims to leverage the generated fine-grained prototypes as guidance to optimize features by focusing on interclass and intraclass distribution separability, thereby characterizing the phenological diversity of crops. Second, to more accurately assess the phenological differences between crops, the comprehensive evaluation of both distance and shape similarity of phenological stages within the model is conducted, and a new temporal metric is proposed to enhance the model's discriminative capability. Finally, the phenological invariant feature learning (PIFL) module is introduced to enhance the model's generalization capability in diverse scenarios. By aligning multiple phenological distribution features within crops, PIFL extracts phenological invariant features to address distribution shifts. The main contributions include the following three points.

1) A time-series-based MDPL framework is proposed to address the critical challenge of phenological diversity in large-scale crop mapping. This is achieved by explicitly capturing inherent phenological sub-distributions within crop growth cycles from time-series remote sensing imagery.
2) FPG and POP modules are proposed to learn multiple time-series prototypes using the proposed temporal metric, thereby optimizing a feature space that achieves both intercrop discriminability and intracrop diversity. PIFL module that aligns features from different intraclass distributions to learn generalized, phenology-invariant representations.
3) A theoretical proof and extensive experiments conducted on three time-series SAR datasets verify the performance of the proposed method surpasses that of existing state-of-the-art methods.

II. Related Work

A. Traditional Crop Mapping Using Time-Series SAR

Over the past decades, various traditional methods have been developed for crop mapping tasks using time-series SAR images [14]. These methods employ machine learning (ML) algorithms, such as SVM [15], RF [16], and Bayesian models [17], to characterize different phenological stages and spatial distributions of crops, thereby distinguishing crop types. Specifically, these approaches utilize multiple linear polarizations [18] (e.g., VH and VV) from time-series SAR images and apply spatiotemporal feature fusion techniques [19], such as multitemporal backscattering coefficient analysis [20], along with time-series analysis methods to extract phenological features. These features are then used for pixel and parcel level predictions via ML to differentiate various crops. The combination of VH and VV polarization provides a detailed insight into crop characteristics [21], enhancing model accuracy. Nonetheless, manual feature selection relies heavily on the expertise of domain specialists and is vulnerable to the noise present in time-series SAR data.

B. DL-Based Crop Mapping Using Time-Series SAR

In recent years, DL-based methods have garnered significant attention due to their powerful capability for automatic feature extraction [22], [23]. In crop mapping, convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are commonly used to capture spatio-temporal features. Specifically, CNNs can efficiently learn spatial context representations through convolutional kernels, enabling end-to-end classification or semantic segmentation tasks in large-scale SAR images [24], [25]. For instance, Sun et al. [26] employ the 1D-CNN architecture to extract information from time-series data, interpreting the growth morphological changes of crops based on NDVI time profiles in a given land area. Sabir and Kumar [27] then develop a sequential 1D-CNN model architecture for medicinal Psyllium Husk crop mapping, demonstrating CNN's effectiveness in time-series crop classification. Gallo et al. [28] further use a fully convolutional 3-D CNN to conduct semantic segmentation of S2 images. With the

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3-D-convolution kernel, they simultaneously extract temporal and spatial features from the input time-series, thus dynamically generating short-term maps along the seasonal direction and enhancing classification accuracy. Luo et al. [29] and Guo et al. [30] combine a stacked autoencoder (SAE) and a 1-D 1D-CNN to innovatively design a convolutional autoencoder neural network (CAENN), thereby exploring the potential of a hybrid CNN-based architecture in crop mapping tasks. However, the convolutional operations in CNNs are inherently local, primarily focusing on features within limited temporal windows. It is challenging for CNN to capture long-range dependencies between phenological stages. Due to significant advancements in handling long time-series data, RNNs, particularly long short-term memory (LSTM) networks and their bidirectional variants (Bi-LSTM), have been widely adopted in crop mapping. Lin et al. [31] enhance the application of LSTM in rice mapping by utilizing Multitask Learning with time-series SAR data from Sentinel-1. Sun et al. [32] design a DB-BiLSTM model that allows for independent learning of multiple features and uses time-series data from Sentinel-1 for high-precision crop mapping. LSTM also exhibits limitations, primarily manifested in the loss of inadequate modeling of long-distance dependencies [33].

The Transformer and its self-attention mechanism [34] have been widely adopted across various fields, due to their ability to effectively capture long-distance dependency information. Rußwurm and Körner [35] are the pioneers in incorporating attention into time-series remote sensing crop classification. Following this, Garnot and Landrieu [36] introduce the U-TAE architecture, which merges the spatial UNet-based structure with TAE to attain pixel-level crop mapping through semantic segmentation. To address the issue of cross-time zone classification in time-series crop mapping, Nyborg et al. [37] introduce thermal positional encoding (TPE). This innovation enhances the generalization capability of the model by modifying the transformer's positional encoding. Han et al. [38] put forward a spatio-temporal multilevel attention (STMA) model for crop mapping. This model outperforms traditional convolutional models regarding accuracy and generalization ability, enabling efficient crop mapping across multiple datasets. The recent emergence of the Mamba architecture has introduced a promising new paradigm for sequential modeling. Mamba-based models present a compelling alternative to Transformers and RNNs for handling long-range dependencies in time-series data. The Mamba-STFM model [39] shows the performance of Mamba in spatiotemporal fusion tasks, which learn long-range contextual dependencies to generate high-resolution remote sensing imagery. Similarly, ChangeMamba [40] uses a spatiotemporal state space model for precise change detection, highlighting its strength in modeling the complex temporal dynamics between multitemporal images. These works demonstrate the necessity of developing more powerful backbone networks for feature extraction. However, existing methods often overlook the phenological diversity within the time-series data. The oversight consequently makes learning a unified phenological feature representation with invariance challenging for large-scale land mapping tasks, resulting in the generalization ability of these methods still falling short.

C. Prototype Learning in Remote Sensing Images

With the continuous advancement of computer theory and technology, prototypes have been widely used in cognitive theories such as category representation, category learning, and classification [41]. Serving as basic semantic abstract embeddings, class prototypes facilitate the tight clustering of samples within a class and the effective separation between classes, making them widely applicable in numerous fields. Chen et al. [42] present an innovative approach to improve semantic segmentation performance through a multi-branch network architecture guided by category prototypes, which effectively enforces the generation of more compact and discriminative feature representations for each semantic class, thereby enhancing the model's robustness. Furthermore, Tang et al. [43] develop a prototype-guided class-level learning algorithm for remote sensing scene classification. Jiang et al. [44] propose a prototype contrastive adaptation method to align class-centric distributions. Zhou et al. [45] suggest learning multiple prototypes for each category, shaping the pixel embedding space by optimizing the arrangement between embedded pixels and anchored prototypes to achieve dense prediction at the image level. In image processing, modeling each category as a single-prototype representation has been commonly used for classification. However, a fundamental challenge in crop mapping lies in the phenological diversity that results in subclasses within crops. Using a single-prototype approach may lead to the misclassification of different crops due to the coarse prototype space. Moreover, most prototype-based methods are primarily developed for remote sensing images rather than time-series remote sensing images. Therefore, it is essential to investigate a prototype-based method specifically for remote sensing time-series data.

III. Methodology

A. Overview

To further enhance the performance of remote sensing crop mapping and improve adaptability to phenological diversity scenarios, a novel MDPL framework is proposed based on multidistribution time-series prototypes learning from a distribution perspective. Fig. 2 shows the proposed framework, which consists primarily of two steps. The first step involves utilizing the fine-grained prototypes generation (FPG) module and the POP module to achieve fine-grained feature space modeling for time-series images with diverse phenological characteristics. The second step uses the phenological-invariant features learning (PIFL) module to learn phenological invariant features based on the fine-grained modeling generated by the first step. The framework alternates between fine-grained prototypes modeling (Step 1) and invariant feature learning (Step 2) through iterative training, ultimately generating classification results via a discriminative classifier trained on the aligned feature space.

In this work, we adopt STMA's feature extractor [38] as our feature extractor F, given its robust temporal feature extraction capability. In the MDPL framework, the input time-series data is a 4-D tensor denoted as $X \in \boldsymbol{R}^{T\times C\times H\times W}$, where T represents the sequence length, C indicates the number of polarization channels, and $H\times W$ indicates the spatial image size. The input

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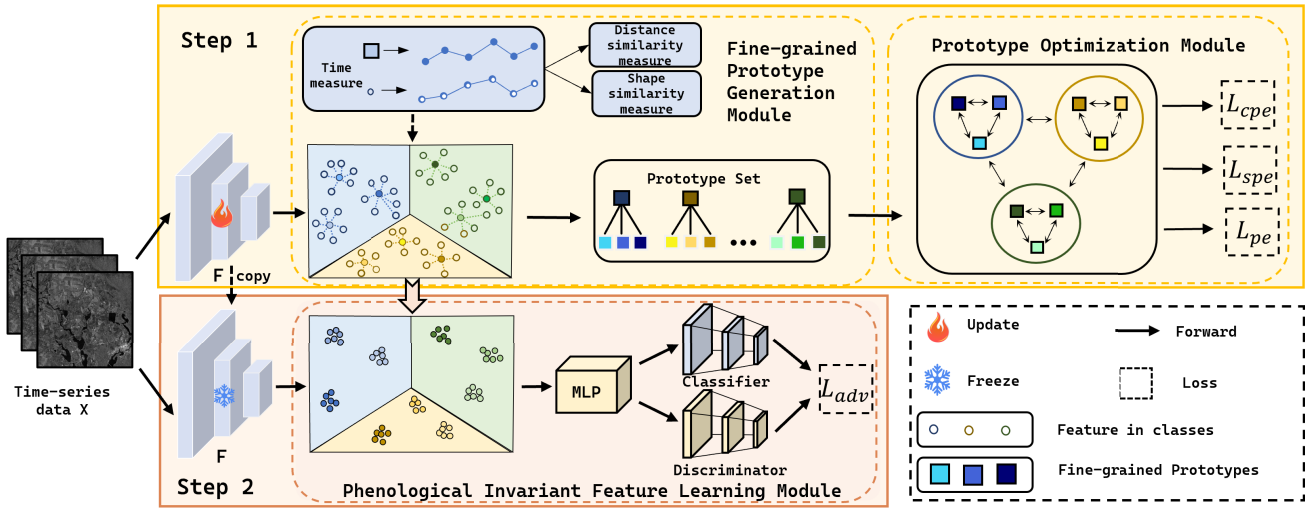

Fig. 2. Overall framework of the proposed MDPL. The loss L_{cpe}, L_{spe}, and L_{pe} are used for constructing a temporal multidistribution feature space, while the loss L_{adv} are used for learning phenology-invariant features.

X is fed into the feature extractor F to obtain the pixel-level feature map $Z = F(X)$, $Z \in \boldsymbol{R}^{D\times H\times W}$, with D representing the temporal feature dimension. In Sections III-B and III-C, the two steps of the MDPL are introduced in detail, along with the theoretical proof of MDPL's enhancement of generalization capability.

B. Fine-Grained Prototypes Modeling

1) Fine-Grained Prototypes Generation Module: In the first step, potential intraclass distributions are captured through fine-grained prototypes learning within the FPG module. Specifically, the FPG module uses a clustering algorithm to learn multiple time-series prototypes to capture the potential phenological distribution of each crop. Additionally, a temporal metric is proposed for crop time-series features to perceive subtle differences in phenological characteristics.

First, multiple time-series prototypes are assigned to each category to establish a fine-grained prototypes set $\{p_{c,k} \in \boldsymbol{R}^D\}_{c,k}^{C,K}$, where C represents the crop categories, K denotes the predetermined number of potential distributions, and $p_{c,k}$ indicates the prototype of the kth distribution in the cth class. Initialization is done by randomly selecting K samples from each category C as initial prototypes.

Then, an unsupervised clustering approach is employed [45], using time-series prototypes as cluster centers to discern the differences between the features and the prototypes. The distribution label of each pixel can be inferred by retrieving the nearest prototype between z_i and $p_{c,k}$ in the following equation:

$$\hat{c}_z = \underset{(c,k)}{\operatorname{argmin}} \left\{\operatorname{dist}\left(z_i, p_{c,k}\right)\right\}_{c,k=1}^{C,K} \tag{1}$$

where dist is a distance metric function, $z_i \in Z$ denotes the feature vector at the ith pixel position in the feature map Z. Moreover, the categories of C are known in this process, and our objective is to obtain potential multiple distribution labels for each crop through fine-grained prototypes. By using distance measurements, the distribution label $\hat{c}_z \in (1, \ldots, k)$ for each feature can be obtained.

To more accurately characterize phenological differences among crops, we proposed a new temporal metric dist in (2), where a smaller value indicates higher phenological similarity. This metric integrates two complementary components: cosine similarity [46] and shape context similarity SCS [47]. These components jointly evaluate similarity from the perspectives of feature direction and magnitude. The motivation for this design is that, although cosine similarity effectively measures the directional consistency between feature vectors, it is insensitive to magnitude variations. As a result, crops with distinct phenological evolution patterns may still exhibit high cosine similarity, leading to potential misclassification. To overcome this limitation, SCS is introduced as a complementary measure that captures amplitude-related variations in temporal evolution from a shape-based perspective, thereby providing a more comprehensive and discriminative representation of crop phenological

$$\text{dist} = \sqrt{d_{\cos}^2 + \alpha\left(1 - \text{SCS}\right)^2} \tag{2}$$

$$\text{SCS}\left(p_{c,k}, z_i\right) = \frac{1}{(t-1)} \sum_{j=1}^{t} \frac{\left(p_{c,k}^j - \mu_{p_{c,k}}\right)\left(z_i^j - \mu_{z_i}\right)}{\sigma_{p_{c,k}} \sigma_{z_i}} \tag{3}$$

where $\mu_{p_{c,k}}$ and μ_{z_i} represent the mean values of the feature vectors $p_{c,k}$ and z_i, while $\sigma_{p_{c,k}}$ and σ_{z_i} denote the standard deviations of feature vectors $p_{c,k}$ and z_i. A higher SCS value indicates a greater shape similarity between the two tensors. The parameter α is a hyperparameter that controls the balance between the cosine distance and shape similarity within dist, where $d_{\cos}$ represents the cosine distance.

To ensure that the prototypes effectively capture feature information from all samples while maintaining smooth updates during training and avoiding suboptimal solutions caused by abrupt changes, we did not directly use the computed feature centroids as new prototypes. Instead, we adopted an exponential moving average (EMA)-based strategy [48] for prototype updating. This approach stabilizes training by suppressing drastic fluctuations in prototype values, thereby

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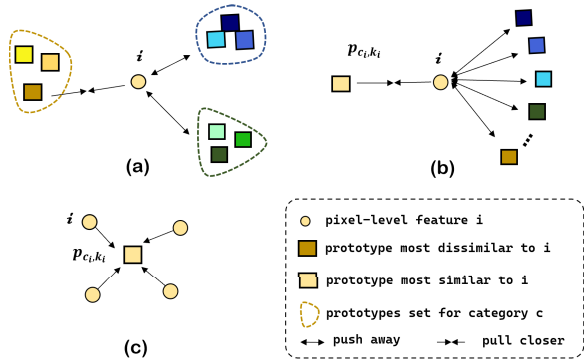

Fig. 3. Illustrations of POP. (a) Interclass relation. (b) Intraclass relation. (c) Intradistribution compactness.

avoiding suboptimal solutions, all while operating without extra parameters and leveraging online clustering results

$$p_{c,k} = \beta p_{c,k} + (1-\beta)\,\bar{z}_{c,k} \tag{4}$$

where $\beta \in (0,1)$ represents the update coefficient, while $\bar{z}_{c,k}$ denotes the feature vector of the kth distribution within class c. It represents the centroid of the corresponding features, the mean feature vector after L_2 normalization.

2) POP Module: To better represent the phenological diversity relationships in the time domain of remote sensing in the first step. This module analyzes from three perspectives. First, to optimize interclass separability L_{cpe} in (5), prototypes of different classes should be optimally dispersed from each other, thereby minimizing confusion between crops and mitigating the impact of similar phenological characteristics. Second, to optimize intraclass separability L_{spe} in (6), multiple prototypes within the same class should maintain an appropriate distance from each other without compromising interclass separability, thereby maximizing the distribution within the class. Through the joint optimization of L_{cpe} and L_{spe}, the POP module effectively enhances the intraclass separability of features while maintaining strong interclass discriminability. Finally, to optimize clustering compactness L_{pe} in (7), the clusters surrounding each prototype should be as close as possible, thereby enhancing the discriminative ability between different prototypes.

Equation (5) is derived as an improvement over the cross-entropy loss, where $\Phi_{z,c} = \min\{\text{dist}(z, P_{c,k})\}_{k=1}^{K}$ represents the distance between z and the least similar prototype within its corresponding category c. As shown in Fig. 3(a), for any given feature, enforces that the distance between a feature and the most dissimilar prototype within its own category is significantly smaller than the distance to any prototype from other categories, which helps maintain clear interclass separability

$$L_{cpe} = -\log \frac{\exp\left(-\Phi_{z,c}\right)}{\exp\left(-\Phi_{z,c}\right) + \sum \exp\left(-\Phi_{z,c'\neq c}\right)}. \tag{5}$$

As shown in Fig. 3(b), for any given feature, enforce its proximity to the most similar prototype within its class while simultaneously distancing it from fine-grained prototypes of other classes (including other prototypes within the same class), thereby preserving the separability of intraclass diversity distributions. Therefore, in (6), $\Theta_z = \max\{\text{dist}(z, P_{c_z,k})\}_{k=1}^{K}$, which represents the maximum distance between feature z and the K prototypes of its corresponding class c_z, where c_z denotes the class of z, and $p_{c_z,k}$ denotes the kth prototype of class c_z. Additionally, we define a new set of negative samples $P_z^- = \{p_{c,k}\}_{c,k=1}^{C,K} / p_{c_z,k_z}$, which are used to construct contrastive pairs with the positive sample p_{c_z,k_z}, where p_{c_z,k_z} is the prototype closest to z, k_z denotes the prototype label corresponding to feature z, and P_z^- contains all other prototypes except this closest one

$$L_{spe} = -\log \frac{\exp\left(-\Theta_z\right)}{\exp\left(-\Theta_z\right) + \sum_{p^- \in P^-} \text{dist}\left(z, p^-\right)}. \tag{6}$$

In intuitive terms, L_{spe} can reduce the distance between each feature and its assigned prototype p_{c_z,k_z} (positive prototype), while increasing the distance between each feature and other prototypes p^- that belong to the same category (negative prototypes).

To enhance the compactness of clusters centered around each prototype, features should be tightly aggregated around their corresponding prototypes. As shown in Fig. 3(c)

$$L_{pe} = -\sum_{i=1}^{n} \|z_i - p_{c,k}\|^2. \tag{7}$$

In the POP module, the total loss for POP can be expressed as follows:

$$L_p = L_{pe} + rL_{cpe} + (1-r)\,L_{spe} \tag{8}$$

where r denotes the weight coefficients.

Feature separation is achieved through a loss-guided prototype, which, respectively, optimizes the overall learning process from three different perspectives: intraclass relationships, interclass relationships, and clustering compactness. Consequently, by learning multiple prototypes for each category, the phenological diversity relationship can be comprehensively described within the temporal feature space, thereby enhancing the representational capacity of the feature space, significantly improving the separability of crops, and reducing the risk of misclassification.

C. Invariant Feature Learning Based on Fine-Grained Prototypes

In the second step, based on the learned intraclass diversity distribution feature space, the PIFL module is utilized to achieve feature alignment among multiple intraclass distributions, thereby enabling the learning of phenological invariant features. Specifically, F is first frozen to capture discriminative features within the temporal-spatial domain. These features are then projected through an MLP block, followed by the design of a classifier for crop category prediction and a discriminator for identifying the corresponding prototype distribution. Both the classifier and the discriminator adopt two standard 3×3 convolutional blocks, followed by a 1×1 convolutional block to post-process the mapped features, producing the final classification map and distribution label predictions.

In the PIFL module, the classifier is responsible for accurately assigning each pixel-level feature in the input image

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to its corresponding crop category, while the discriminator aims to distinguish input features based on their associated prototype distributions. A gradient reversal layer (GRL) [49] is utilized to update the loss of the distribution discriminator. It forces the alignment of subdistribution features from different distributions and achieves phenology-invariant feature learning. The specific formula is as follows:

$$F_{\text{mlp}} = \text{MLP}\left(F\left(x\right)\right) \tag{9}$$

$$L_{\text{adv}} = l_{\text{ce}}\left(C_c\left(\text{mlp}\right), c\right) + l_{\text{ce}}\left(D_k\left(R_\lambda\left(F_{\text{mlp}}\right)\right), k\right) \tag{10}$$

where l_{ce} represents the standard cross-entropy loss, C_c denotes the class classifier, D_k represents the distribution discriminator, and R_λ is the gradient reversal layer controlled by the hyperparameter λ.

The entire workflow is shown in the Algorithm. 1.

Algorithm 1 Framework of MDPL

Input: X: labeled samples
Output: Trained model F

```
Randomly initialize F and prototypes set
while epoch ≤ Epoches do
    Sample batch x_i from X
    /* this is for first step */
    for x_i in X do
        Calculate dist with Eq. (2)
        Update P with Eq. (4)
        Update pseudo-label of class distribute
        Calculate L_p with Eq. (8)
    end for
    Update F
    /* this is for second step */
    for x_i in X do
        Calculate L_adv with Eq. (10)
    end for
    Update MLP, C_c and D_k
    epoch = epoch + 1
end while
```

D. Theoretical Insight

In conventional approaches [50], the goal is to maintain interclass separability while also ensuring intraclass similarity, even to the extent of minimizing the differences within intraclass distributions. However, in the POP module, the aim is to preserve intraclass separability, even to the extent of maximizing intraclass distribution differences. Additionally, in the PIFL module, an alignment strategy is enforced among these multiple distributions. It may cause confusion, as it diverges from common practices. This section introduces an analysis of the generalization error bound [51], aimed at elucidating the objectives and the necessity of this approach, as detailed below.

Proposition: Let I represent a time-series space with phenological diversity in remote sensing scenarios and $\mathcal{H}$ correspond to the hypothesis within this space. Let $\mathcal{Q}$ and $\mathcal{P}$ be distributions in I, where $\mathcal{Q}$ is often expressed as the unknown distribution during testing, while $\mathcal{P}$ represents the learnable distribution in the known data. Due to the diversity of phenological characteristics of crops, there exists $\mathcal{P} = \{\varphi_k \mathcal{P}_k\}_{k=1}^{K}$, where $\mathcal{P}_k$ denote the potential subdistribution over $\mathcal{P}$. The coefficients φ_k are a set of nonnegative numbers satisfying the condition $\sum_k \varphi_k = 1$. Assume that $\mathcal{S}$ is the distribution representation after model training, $\forall S \in \mathcal{P}$, for any $\mathcal{P}_k$ the following holds:

$$d_{\mathcal{H}\Delta\mathcal{H}}\left(\mathcal{P}_k, \mathcal{S}\right) \le \max_{i,j} d_{\mathcal{H}\Delta\mathcal{H}}\left(\mathcal{P}_i, \mathcal{P}_j\right) \tag{11}$$

where $d_{\mathcal{H}\Delta\mathcal{H}}()$ refers to the H divergence.

According to the upper bound of the errors in [51], for any $h \in \mathcal{H}$, the following relation (12) holds, where ε' denotes the error of an ideal joint hypothesis for $\mathcal{Q}$, $\mathcal{P}$, representing the upper bound of the target error. $\varepsilon_{\mathcal{Q}}(h)$ denotes the expected error of a hypothesis $h \in \mathcal{H}$ on the target distribution $\mathcal{Q}$, while $\varepsilon_P(h)$ represents the expected error of the same hypothesis on the source distribution $\mathcal{P}$. Equation (12) highlights the significance of learning domain-invariant representations in multidistribution scenarios

$$\varepsilon_{\mathcal{Q}}\left(h\right) \le \varepsilon' + \varepsilon_{\mathbb{P}}\left(h\right) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}\left(\mathcal{Q}, \mathcal{P}\right). \tag{12}$$

Therefore, it can be inferred from (11) to (12) that for any $h \in \mathcal{H}$ [52], have (13)

$$\varepsilon_{\mathcal{Q}}\left(h\right) \le \varepsilon'' + \sum_k \varphi_k \varepsilon_{\mathcal{P}_k}\left(h\right) + \frac{1}{2}\min_{\mathcal{S}\in\mathcal{P}} d_{\mathcal{H}\Delta\mathcal{H}}\left(\mathcal{S}, \mathcal{Q}\right) + \frac{1}{2}\max_{i,j} d_{\mathcal{H}\Delta\mathcal{H}}\left(\mathcal{P}_i, \mathcal{P}_j\right) \tag{13}$$

where ε'' represents the ideal joint intrinsic error, $\varepsilon_{\mathcal{P}_k}(h)$ denotes the error of hypothesis h under distribution $\mathcal{P}_k$, which measures the interdistribution distance.

The above (13) describes the training error of the unknown test distribution on the model [53]. The first term, ε', which represents a negligible constant, is typically disregarded. The second term, $\sum_k \varphi_k \varepsilon_{\mathcal{P}k}(h)$, appears in almost all methods. Simply put, the distributional discrepancy is minimized using the cross-entropy loss. Therefore, the primary focus of optimization in this study lies in the latter two terms.

Term $(1/2)\max_{i,j} d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{P}_i, \mathcal{P}_j)$ is used to measure the maximum discrepancy between data distributions. Therefore, it is expected that the distributional divergence can be minimized through class-invariant representation learning, reducing this value as much as possible.

For the third term, $(1/2)\min_{\mathcal{S}\in 0} d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{S}, \mathcal{Q})$ explains why subclass distributions need to be separated and their distances increase. Here, $\mathcal{Q}$ represents an unseen distribution, which can be simply understood as the test domain. Due to the complexity of temporal-spatial dynamics in remote sensing, it is challenging to capture all phenological traits of crops, making it difficult to comprehensively evaluate the spatial distribution of $\mathcal{Q}$. To minimize the third term, expanding S becomes the only viable approach. It corresponds to the POP module, which forces increased distances between subclass distributions through the use of prototypes. This enlarges the distributional space while reducing the gap with the unseen distribution.

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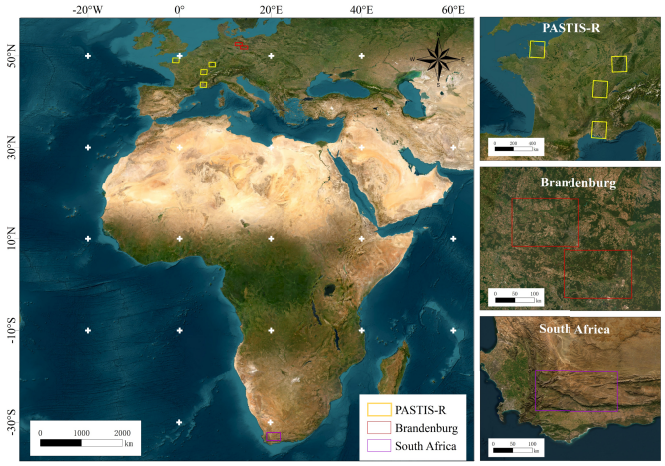

Fig. 4. Location of the study areas in the PASTIS-R, Brandenburg, and South African datasets, collected from France, Brandenburg, and Western Cape, respectively.

TABLE I
PER LAND COVER STATISTICS FOR THE PASTIS-R DATASET

Class ID	Class Name	# Pixels	% Pixels
0	Bakground	15907494	39.91%
1	Meadow	7289116	18.29%
2	Soft winter wheat	2733960	6.86%
3	Corn	3451442	8.66%
4	Winter barley	810895	2.03%
5	Winter rapeseed	705329	1.77%
6	Spring barley	254693	0.64%
7	Sunflower	395551	0.99%
8	Grapevine	1108797	2.78%
9	Beet	324825	0.81%
10	Winter triticale	294045	0.74%
11	Winter durum wheat	419760	1.05%
12	Fruits, vegetables, flowers	373283	0.94%
13	Potatoes	111600	0.28%
14	Leguminous fodder	635561	1.59%
15	Soybeans	420060	1.05%
16	Orchard	412080	1.03%
17	Mixed cereal	207035	0.52%
18	Sorghum	166966	0.42%
19	Other	3839780	9.63%
	Total	39862272	100%

IV. EXPERIMENT AND ANALYSIS

A. Experimental Dataset

1) PASTIS-R [54]: The PASTIS-R dataset is the extended version of PASTIS [36], is a labeled dataset built specifically for agricultural remote sensing analysis, and a benchmark dataset for crop mapping based on Sentinel-1 and Sentinel-2 satellite images. As shown in Fig. 4, covering the agricultural area of France. The dataset contains 2433 time-series samples across 18 crops. PASTIS-R was collected from September 2018 to November 2019, and the Sentinel-1 sensor acquired time-sequence SAR in three different modes, including VV, VH and VV/VH, providing a wealth of polarization information. It is worth noting that this dataset is highly unbalanced, with an imbalance ratio of 56.8, as shown in Table I.

2) Brandenburg [38]: The Brandenburg dataset contains high-quality remote sensing and ground-based crop classification data, focused on the German state of Brandenburg and designed to support research related to common agricultural policy. The dataset provides 10-m resolution, 12-day revisit period Sentinel-1 images with a size of 3731 × 5095 pixels, which are used to precisely demarcate plot boundaries and identify crop categories. The data is spatially and temporally separated, ensuring the generalization ability of the model across different years and regions. The region was fixed at 24 × 24km, covering 15 major crops (e.g., maize, wheat, rapeseed) and other land use types (e.g., fallow land), as shown in Table II.

TABLE II
PER LAND COVER STATISTICS FOR THE BRANDENBURG DATASET

Class ID	Class Name	# Pixels	% Pixels
0	Bakground	12145780	60.63%
1	Maize	1213098	6.10%
2	Wheat	2986929	15.28%
3	Grassland	2351940	12.28%
4	Peanut	96619	0.45%
5	Potato	79992	0.40%
6	Residue	25149	0.13%
7	Fallow	11176	0.06%
8	Rapeseed	613027	3.12%
9	Vegetable	7765	0.04%
10	Legume	113850	0.57%
11	Herb	1059	0.01%
12	Orchard	2786	0.01%
13	Flower	1805	0.01%
14	Sugar beet	5015	0.03%
15	Other	4810	0.03%
	Total	19660601	100%

TABLE III
PER LAND COVER STATISTICS FOR THE BRANDENBURG DATASET

Class ID	Class Name	# Pixels	% Pixels
0	Bakground	15280652	24.91%
1	Lucerne/Medics	6392079	10.42%
2	Planted pastures	4862998	7.93%
3	Fallow	3937965	6.42%
4	Wine grapes	3311925	5.40%
5	Weeds	2254971	3.68%
6	Small grain grazing	7213470	11.76%
7	Wheat	13869891	22.61%
8	Canola	2053353	3.35%
9	Rooibos	2164392	3.53%
	Total	61341696	100%

3) South Africa [55]: The South African dataset was developed for the Radiant Earth "Spot the Crop Challenge" to use remote sensing to classify crops in the Western Cape Province. The dataset contains Sentinel-1 satellite time-series data collected between 2017 and 2018. Two-season or multi-season cropping systems caused by tropical climates increase the difficulty of crop identification and location. The dataset contains a total of 2650 training samples and 1137 testing samples, covering 9 crop categories, as shown in Table III, and recording time-series SAR image observation data for VV and VH modes.

B. Implementation Details

To unify the data settings, it is necessary to perform initialization processing on the dataset. First, the mean and standard deviation for each polarization pattern are calculated, using a z-score [56] to normalize each polarization direction. After that, the sliding window is used to crop the image into a separate patch 64 × 64 to unify the data input. Finally, 60% of the patches were randomly selected as the training set, 20% as the verification set, and 20% as the testing set.

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TABLE IV

QUANTITATIVE COMPARISON OF RESULTS FOR DIFFERENT CROP MAPPING METHODS ON PASTIS-R DATASET, BRANDENBURG DATASET, AND SOUTH AFRICA DATASET, WHERE OA, MEAN F1 (MF1), MEAN IOU (MIOU). THE BEST PERFORMANCE OF EACH INDICATOR IS SHOWN IN BOLD

Method	PASTIS-R			Brandenburg			South Africa		
	OA	mIoU	mF1	OA	mIoU	mF1	OA	mIoU	mF1
SST-CLT [57]	78.78%	54.85%	68.03%	87.45%	49.98%	50.72%	64.25%	36.30%	49.04%
3D-Unet [58]	80.80%	56.27%	69.63%	92.40%	60.13%	69.34%	67.46%	38.29%	52.75%
U-TAE [36]	82.74%	57.19%	70.38%	91.14%	55.84%	63.39%	68.56%	38.36%	51.98%
Temp-RNN [37]	83.46%	60.34%	73.92%	91.93%	56.57%	65.82%	70.23%	41.56%	54.15%
SIST-MoCo [59]	82.34%	58.61%	70.76%	92.55%	62.24%	71.42%	71.26%	40.51%	53.45%
3D-M2C [60]	83.65%	59.45%	71.83%	92.58%	62.77%	72.48%	72.24%	41.8%	54.13%
STMA [38]	84.06%	62.67%	72.29%	93.08%	63.81%	73.21%	72.72%	43.85%	57.64%
MDPL (**Our**)	**86.91%**	**65.78%**	**76.82%**	**94.12%**	**64.02%**	**74.73%**	**75.46%**	**48.42%**	**61.36%**

In order to achieve a fair comparison, all experiments were trained on the Pytorch platform using one NVIDIA GeForce 3090 GPU. During the training process, the learning rate of the training strategy based on warm-up increases linearly in the first 20 steps, and then slowly decreases. The initial learning rate is 1e-3. The model updates network parameters based on the Adam optimizer. Batch size is set to 16 and the maximum number of epochs is set to 200.

Referring to [36] and [38], the overall accuracy (OA), mean value of IoU (mIoU) and mean $F1$ score (mF1) are used to evaluate the performance of different methods for remote sensing crop mapping

$$\mathrm{OA} = \frac{\sum_{i=1}^{C} \mathrm{TP}_i}{N} \tag{14}$$

$$\mathrm{mIoU} = \frac{1}{C} \sum_{i=1}^{C} \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i} \tag{15}$$

$$\mathrm{mF1} = \frac{1}{C} \sum_{i=1}^{C} \frac{2 \cdot \mathrm{TP}_i}{2 \cdot \mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i} \tag{16}$$

where C denotes the total number of classes, and TP_i, FP_i, and FN_i represent the numbers of true positives, false positives, and false negatives for the ith class, respectively. N denotes the total number of test samples.

C. Comparison With SOTA

The performance of the proposed MDPL is compared on the PASTIS-R, Brandenburg, and South Africa datasets with five state-of-the-art crop mapping algorithms, namely SST-CLT [57], 3D-Unet [58], U-TAE [36], Temp-RNN [37], SIST-MoCo [59], 3D-M2C [60] and STMA [38]. The parameter settings of the compared algorithms are consistent with those in the original article. Table IV shows the performance of various methods. Notably, our MDPL method demonstrates a superior advantage over other models across all three datasets.

1) Performance Comparison on the PASITS-R Dataset: As shown in Table IV, MDPL achieved an OA of 86.91%, an mIoU of 65.78%, and a mF1 of 76.82%, all surpassing those of the comparison methods. The notable increase in the mF1 score indicates that MDPL effectively identifies positive samples, enhancing both accuracy and recall. Compared with the pre-training-based SITS-MoCo method, MDPL achieves a 4.57% improvement in OA. When compared with the multi-scale attention-based STMA method, it attains gains of 3.11% and 4.53% in mIoU and mF1, respectively. These results demonstrate that MDPL can more effectively distinguish crop differences at a fine-grained level and provide a more comprehensive representation for crop mapping. Meanwhile, Fig. 5 illustrates the confusion matrices of all methods on the PASITS-R dataset, which represent the mapping accuracy for each crop. In the PASTIS-R dataset, most models achieved over 90% classification accuracy for several sample categories, including meadow, soft winter wheat, corn, winter rapeseed, and grapevine. However, the classification performance for sorghum was generally poor, with most samples being misclassified as corn, likely due to their similar phenological characteristic. Notably, the MDPL model demonstrates a better ability to capture the internal differences in sorghum phenology, achieving the highest classification accuracy among the models assessed.

2) Performance Comparison on the Brandenburg Dataset: In the Brandenburg dataset, MDPL exhibits an average performance improvement of approximately 2.92% in OA, 6.95% in mIoU, and 10.24% in mF1 compared to other methods in Table IV. While the improvement in OA may not be substantial, the significant increase in mF1 highlights MDPL's robust adaptability in remote sensing scenarios characterized by uneven category distribution. As shown in Fig. 6, it can be observed that the Brandenburg dataset produces results similar to those of the PASTIS-R dataset, where high classification accuracy was achieved across most classes. Simultaneously, the SST-CLT, 3D UNet, and U-TAE are prone to misclassifying crops as Maize, Wheat, and Grassland. The underlying reason for this may be that these models exhibit a preference for the majority classes and are unable to address the issue of class imbalance. In contrast, as illustrated in Fig. 6, both STMA and MDPL performed exceptionally well under conditions of class imbalance. STMA exhibited higher accuracy for herb and sugar beet, while MDPL maintained a more balanced and stable performance overall.

3) Performance Comparison on the South Africa Dataset: In the South Africa dataset, overall performance across all methods is relatively modest due to the region's complex climatic conditions. Nevertheless, MDPL maintains a competitive edge, the corresponding OA, mIoU, and mF1 values are 75.46%, 48.42%, and 61.36%, compared to the second-best

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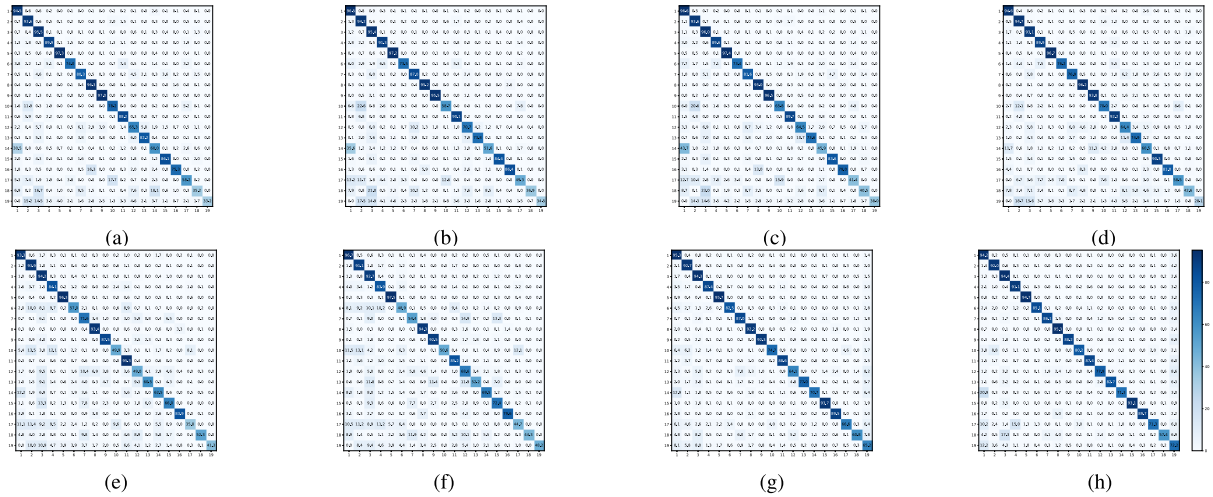

Fig. 5. Confusion matrices obtained on the PASTIS-R dataset by different comparison methods. (a) SST-CLT. (b) 3D UNet. (c) U-TAE. (d) TPE-RNN. (e) SITS-MoCo. (f) 3D-M2C. (g) STMA. (h) MDPL.

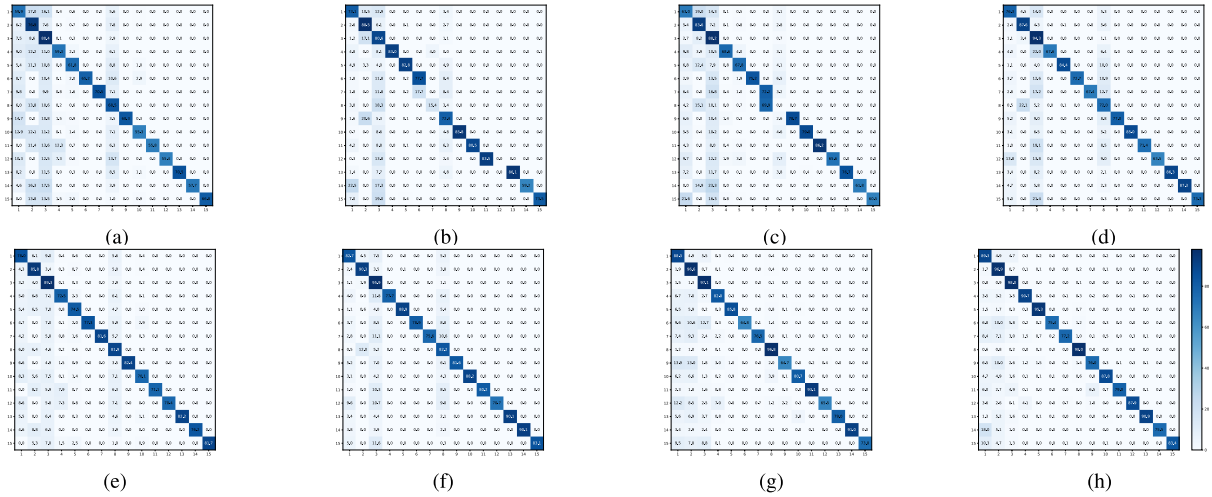

Fig. 6. Confusion matrices obtained on the Brandenburg dataset by different comparison methods. (a) SST-CLT. (b) 3D UNet. (c) U-TAE. (d) TPE-RNN. (e) SITS-MoCo. (f) 3D-M2C. (g) STMA. (h) MDPL.

algorithm, STMA, it demonstrates improvements of 2.14% in OA, 4.57% in mIoU, and 3.72% in mF1. As shown in Fig. 7, MDPL achieved optimal scores across multiple categories, showcasing its superiority in addressing the significant internal phenological differences present. These results collectively affirm the superiority of MDPL in the context of crop mapping, particularly in challenging environments.

D. Visualization Analysis

Conduct a visual analysis experiment using T-SNE to evaluate the effectiveness of MDPL in scenarios with phenological diversity. A comprehensive analysis was conducted using the output from the last network layer as input features for T-SNE [61] analysis, comparing 15 crop classes within the Brandenburg dataset. For each class, a systematic selection of 100 feature points was made to ensure a balanced representation, as illustrated in Fig. 8. It can be observed that, compared to other methods, MDPL has clearer classification boundaries, while due to the feature alignment performed by PIFL, the features of the same crop exhibit more advantageous aggregation, whereas other methods may result in multiple clusters for a single category. In addition, the incorporation of the POP module plays a crucial role in enhancing feature separation. This module enables the feature maps of different crops to maintain relative independence in the feature space, thereby improving classification accuracy.

Meanwhile, the crop mapping results corresponding to different comparison methods on the Brandenburg dataset are visualized, as shown in Fig 9. Based on a resolution of 10 m, crop mapping for two plots was displayed at a scale of 500 $\times$ 500 pixels. It can be observed that MDPL still demonstrates excellent performance in large-scale crop mapping, even achieving high-precision classification for minority classes such as linseed and residue. However, MDPL has limitations at the boundaries of the mapped plots, as it struggles to effectively separate the intermediate background areas between adjacent plots.

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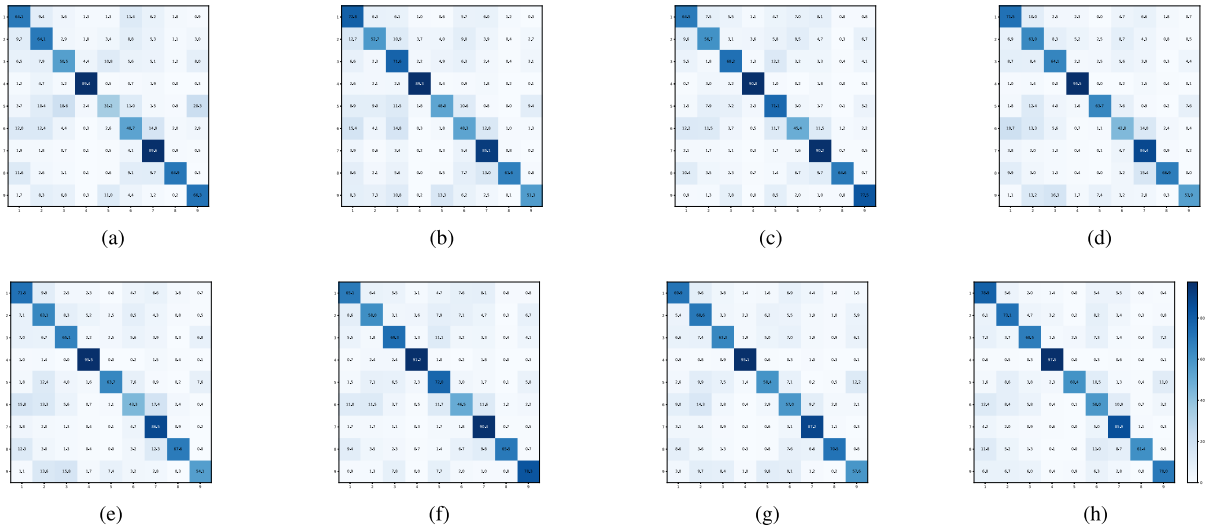

Fig. 7. Confusion matrices obtained on the South Africa dataset by different comparison methods. (a) SST-CLT. (b) 3D UNet. (c) U-TAE. (d) TPE-RNN. (e) SITS-MoCo. (f) 3D-M2C. (g) STMA. (h) MDPL.

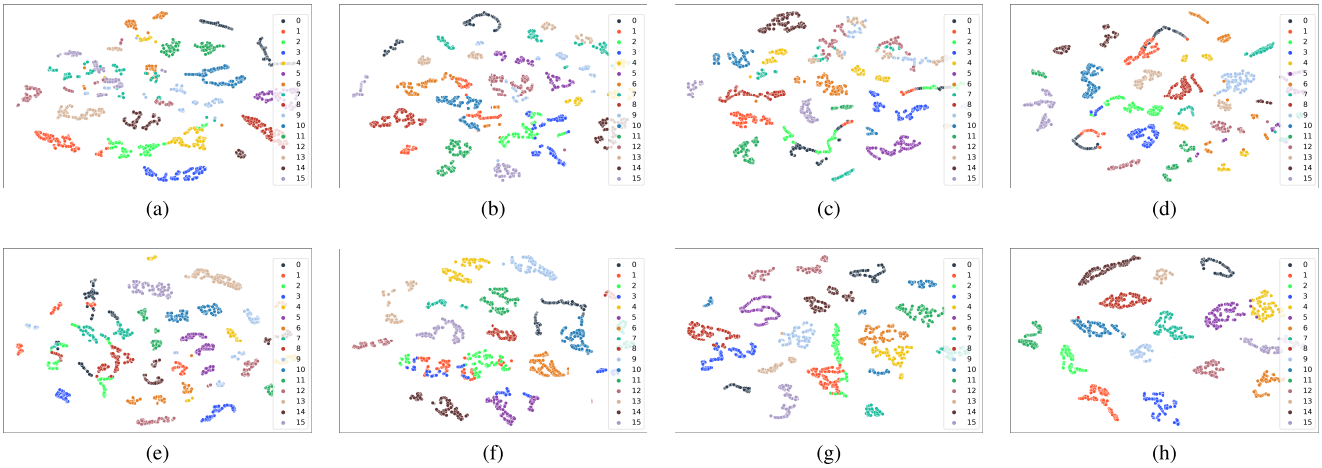

Fig. 8. T-SNE visualization of internal feature representation learned by different crop mapping methods. (a) SST-CLT. (b) 3D UNet. (c) U-TAE. (d) TPE-RNN. (e) SITS-MoCo. (f) 3D-M2C. (g) STMA. (h) MDPL.

The generalization capabilities of various models are also investigated as the proportion of available labeled samples increases. The models are trained using 10%–100% of the data from the PASTIS-R benchmark dataset, with the results depicted in Fig. 10. The model's performance tends to improve with the increasing number of available samples across many metrics. Even with a small sample size, such as 10%, the overall performance of MDPL remains superior to that of other methods. As the data size increases to 60%, the rate of improvement in the model begins to slow down. Compared to STMA, MDPL achieves an increase of 2.85% in OA and 3.93% in mF1. Overall, MDPL outperforms the state-of-the-art models, and these results demonstrate its generalization ability with limited sample sizes.

E. Ablation Study

To verify the effectiveness of the proposed FPG, POP, and PIFL modules, ablation experiments were conducted on the MDPL model, as summarized in Table V. The specific configurations examined in the experiments are as follows.

1) *Baseline:* This configuration includes only the feature extractor and the classification head;
2) *Baseline* + FPG_{cosine} + *POP:* In this setup, cosine distance is utilized as the distance loss for the FPG model;
3) *Baseline* + FPG_{dist} + *POP:* dist is the proposed temporal metric;
4) *Baseline* + FPG_{dist} + *PIFL:* This configuration excludes the POP module;
5) Baseline + FPG_{dist} + POP + PIFL: This represents the complete MDPL model.

Table V presents the results for the Brandenburg dataset. The baseline model achieves OA of 89.14%, mF1 of 66.08%, and mIoU of 57.26%. When only the FPG module, along with the cosine metric function and the POP module, is added to the baseline, significant improvements are observed: OA increases by 2.11%, mF1 by 4.13%, and mIoU by 4.47%.

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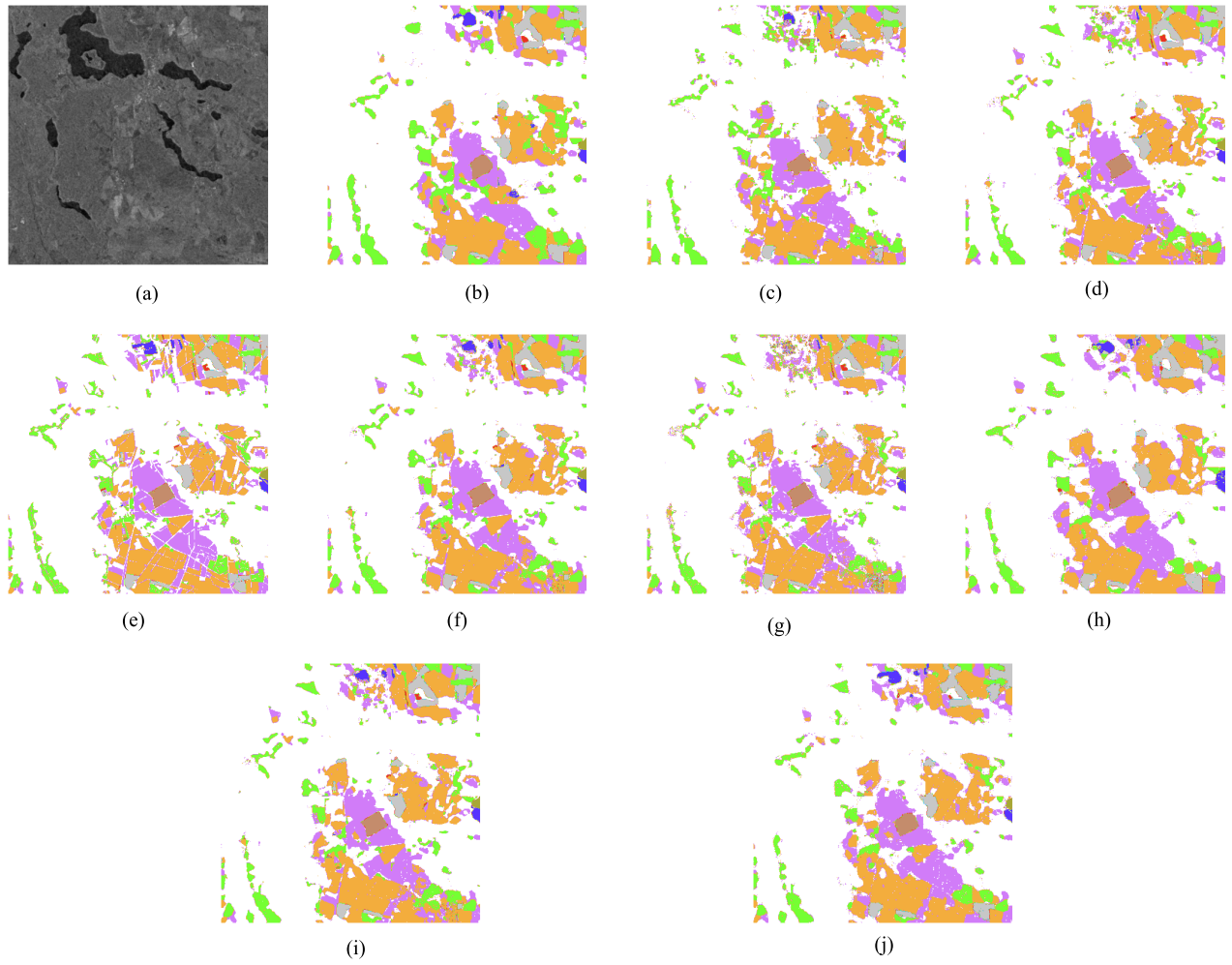

Fig. 9. Crop mapping results of different methods on Brandenburg Sentinel-1 dataset. (a) SAR. (b) SST-CLT. (c) 3D UNet. (d) U-TAE. (e) Ground reference. (f) TPE-RNN. (g) SITS-MoCo. (h) 3D-M2C. (i) STMA. (j) MDPL.

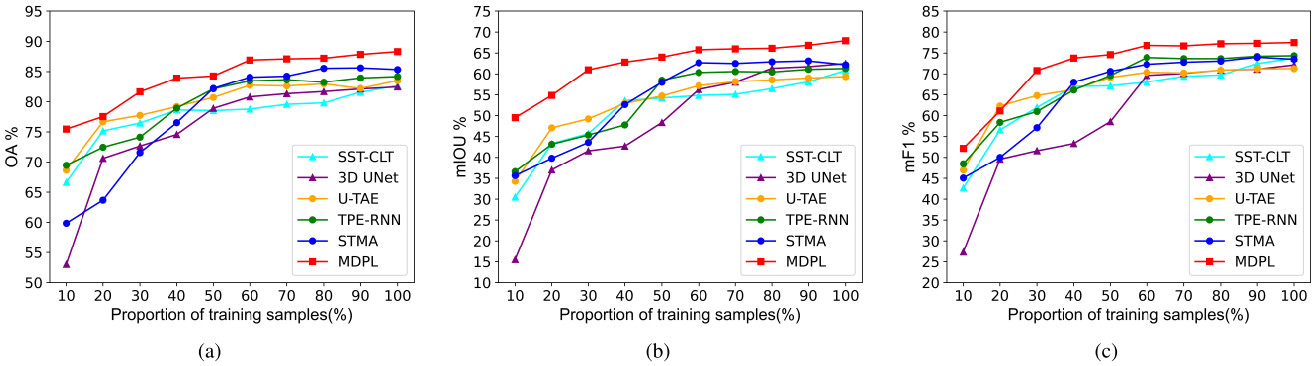

Fig. 10. Evaluation of accuracy for different comparison methods under varying training sample ratios. (a) OA. (b) mIoU. (c) mF1.

These enhancements indicate that the proposed FPG and POP modules effectively identify potential characteristics in the data, improving the separation between category features and modeling the temporal feature space. With the introduction of the SCS distance metric, the model's performance further improves, reaching OA of 93.42% and mF1 of 72.72%. It demonstrates that the dist metric reduces interference between classes while enhancing the boundary separability between

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TABLE V
PER LAND COVER STATISTICS FOR THE BRANDENBURG DATASET

Ablation	OA	mF1	mIOU
Baseline	89.14%	66.08%	57.26%
Baseline+FPG$_{cosine}$+POP	91.25%	70.21%	61.73%
Baseline+FPG$_{dist}$+POP	93.42%	72.12%	62.7%
Baseline+FPG$_{dist}$+PIFL	90.25%	69.29%	60.12%
Baseline+FPG$_{dist}$+POP+PIFL	94.12%	74.73%	64.02%

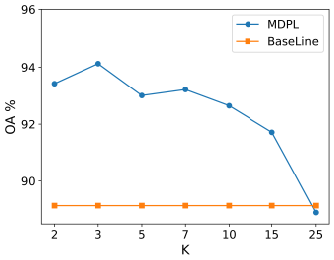

Fig. 11. Impact of the number of prototypes K on model performance.

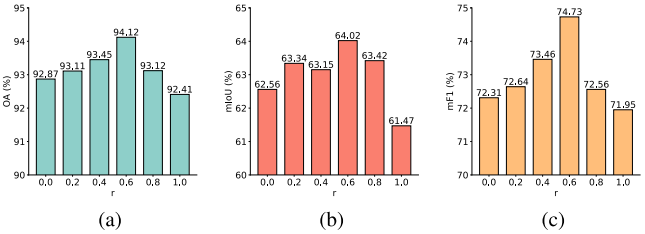

Fig. 12. Influence of the hyperparameter r. (a) OA. (b) mIoU. (c) mF1.

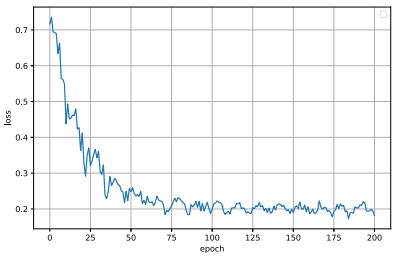

Fig. 13. Convergence results on the PASTIS dataset.

different categories. The inclusion of the PIFL module yields the best performance, with OA reaching 94.12%, mF1 at 74.73%, and mIoU at 64.02%. The PIFL module focuses on learning phenological-invariant features, thereby increasing the model's robustness, and the performance in scenarios with phenological diversity is enhanced. To confirm that MDPL represents the optimal configuration, an experiment was also conducted in which the POP module was removed. It resulted in a significant decline in performance, with OA dropping to 90.25% and mF1 decreasing by 4.94% compared to the full MDPL model. This highlights the crucial role of the POP module in MDPL, as it effectively models time-series feature spaces. Additionally, the POP module aids in separating distances between different category features, expanding the feature space and improving the model's generalization capability.

F. Hyperparameters Discussion

This section provides an analysis and discussion of the impact of hyperparameters on crop mapping accuracy in detail. Our first exploration focuses on the effect of the number of generated prototypes K on the OA, as illustrated in Fig. 11. When $K = 1$, the model degenerates into the baseline method, the results indicate that the best performance is achieved with three prototypes; in addition, a gradual decrease in accuracy is observed as the number of prototypes increases. When $K = 25$, the model's performance degrades significantly, even falling below that of the baseline methods.

For the MDPL, the use of multiple prototypes allows for better capture of the underlying data distribution within each category. However, setting too many prototypes can distort the original data distribution, fragmenting a coherent representation into overly fine-grained clusters, which negatively impacts data representation and increases unnecessary computational overhead. Conversely, when the number of prototypes is too small, the model fails to sufficiently capture the underlying data distributions, resulting in suboptimal performance. Thus, finding the optimal number of prototypes is crucial for balancing data representation and computational efficiency.

TABLE VI
COMPLEXITY ASSESSMENT OF DIFFERENT METHODS

Method	Params (M)	GFLOPs	Inference time (ms)	Train time (h)
SST-CLT	0.97	1000.34	22.12	9.08h
3D-Unet	1.55	277.58	56.77	8.54
U-TAE	1.07	187.02	50.17	9.24
Temp-RNN	0.20	100.52	34.52	8.29
STMA	45.01	36.55	14.35	9.2
MDPL (**Our**)	23.89	95.49	16.99	15.8

We also investigate the optimal value of the hyperparameter r to examine the performance of L_{cpe} and L_{spe} under different weighting ratios. Fig. 12 presents the performance of the proposed methods as r is varied. As shown in Fig. 12, the best performance is achieved when $r = 0.6$. When r decreases, the model tends to overemphasize the differences between prototypes while losing class-specific information, leading to excessively dispersed prototypes and an increased risk of misclassification. Conversely, when r is too large, the contribution of the prototype separation loss L_{spe} becomes weakened. In this case, the model focuses more on distinguishing prototypes of different classes while neglecting the relative relationships among prototypes within the same class. This imbalance may hinder the subsequent learning of phenology-invariant features and ultimately degrade overall performance.

G. Time Complexity And Convergence Analysis

To evaluate the time complexity of MDPL, we calculated the number of parameters, GFLOPs, per-batch inference time, and total training time to assess the algorithm's efficiency, as summarized in Table VI. In addition, we visualized the loss curves of MDPL to examine its convergence behavior, as shown in Fig. 13.

As shown in Table VI, MDPL exhibits a relatively larger number of parameters due to its feature extractor being built

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TABLE VII
EXPLORATORY EXPERIMENTS ON THE BRANDENBURG S2 OPTICAL DATASET

Ablation	OA	mF1	mIOU
SST-CLT [57]	93.67%	45.33%	56.14%
3D-Unet [58]	95.98%	59.32%	72.48%
U-TAE [36]	93.82%	47.54%	58.51%
Temp-RNN [37]	94.31%	54.27%	65.58%
SIST-MoCo [59]	95.06%	63.55%	74.19%
3D-M2C [60]	94.45%	61.04%	73.69%
STMA [38]	95.59%	64.87%	75.3%
MDPL (**Our**)	**95.7%**	**65.95%**	**76.5%**

upon the STMA architecture, which incorporates multiple Transformer modules. In terms of computational cost (measured in GFLOPs), MDPL achieves a lower computational load than STMA and attains the second-best inference speed among the compared methods. However, MDPL shows a noticeably longer training time, primarily because it includes a clustering process during training, which introduces additional computational overhead. This represents one of the current limitations of our approach. Nonetheless, since the clustering process is performed only during training, it does not impact inference efficiency in practical applications such as land-cover mapping.

The results shown in Fig. 13 demonstrate that our method exhibits stable convergence. Although slight fluctuations can be observed, they are primarily caused by variations in sample distributions, which introduce minor perturbations during training. These fluctuations remain within an acceptable range. This phenomenon is common in multidistribution learning approaches and has been widely reported in previous studies [62], [63].

H. Generalization Experiments on Optical Remote Sensing Datasets

Although the proposed MDPL method is primarily optimized for SAR image time series, we conducted additional exploratory experiments on the publicly available optical remote sensing dataset, the Brandenburg S2 dataset, to verify its universality and generalization capability. This dataset is composed of Sentinel-2 optical satellite imagery, whose data modality (optical reflectance) differs substantially from the SAR backscattering coefficients.

As shown in Table VII, without any specific optimization for optical data, MDPL still achieves the most competitive performance on the Brandenburg S2 dataset. Across multiple evaluation metrics, MDPL attains the highest scores, with mF1 and mIoU demonstrating a substantial advantage over other comparative methods. These results clearly indicate that the fine-grained prototype learning mechanism employed by MDPL, along with its capacity for generalized phenological feature learning, is not confined to patterns specific to SAR data but can be effectively transferred and applied to optical remote sensing time series.

V. CONCLUSION

In this article, a time-series remote sensing crop mapping framework based on multidistribution time-series prototype learning is proposed, which enhances crop mapping performance through improved prototype and generalization learning ability. First, multiple time-series prototypes for each class are learned to capture the underlying data distribution and model the complex interclass and intraclass relationships among crops. The approach effectively mitigates mapping errors stemming from the internal growth diversity of crops. In challenging crop growth scenarios, our method effectively separates the spatial distribution of features and optimizes the feature learning process. Additionally, guided by generalization theory, phenological invariant features across multiple latent distributions are learned. The strategy significantly enhances the model's generalization performance, allowing it to maintain excellent results even with a small number of samples. Experiments demonstrate that MDPL achieves the highest accuracy on three time-series SAR mapping datasets across different regions, significantly outperforming other methods and exhibiting strong generalization ability.

Despite the promising results, the proposed MDPL framework primarily models the temporal phenological diversity in a relatively local manner. For future work, motivated by the successes of GNNs in signal processing [64], [65], we plan to explore an end-to-end framework based on Graph Neural Networks [66], [67] to model long-range spatial dependencies and structured relationships between fine-grained prototypes. Specifically, we propose to construct a graph where nodes represent individual prototypes and edges encode their spatial adjacency and feature similarity. This structure would allow the model to propagate information globally, thereby enabling the simultaneous optimization of both the prototype and PIFL modules.

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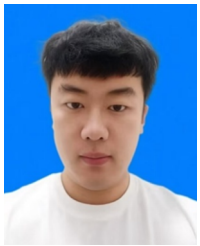

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